Component list

AGV I-PRIME driverless towing vehicle · DRAFT for preliminary review · not for production

Product	AGV I-PRIME driverless towing vehicle
Manufacturer	PT INTI GANDA PERDANA, Jl Pegangsaan Dua Km 1.6, Jakarta Utara, Indonesia
Document	Component list (parts list)
Document number	**IGP-IPR-CL-001**
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Status	DRAFT: for preliminary review, not for production
Related documents	Circuit diagram IGP-IPR-CD-001 · Operation manual IGP-IPR-OM-001
Contents	Sheet "Components": every component on the vehicle, one row per occurrence, with the reference designations (tags) of the circuit diagram. Sheet "Cables": the cables of the circuit diagram.
How to read	Tags as in the circuit diagram: location + device, e.g. +EB-QA1 = main switch in the electrical box. Locations: +BT battery, +EB electrical box, +FR control panel (front), +PD pendant, +VH vehicle.

Revisions

Revision	Change
A · 2026-11-01 · JOY M.	First issue, draft

Components

One row per component, with the tags and page numbers of circuit diagram IGP-IPR-CD-001.

No.	Tag	Component	Manufacturer	Type	Main data	Qty	Diagram pages
Power supply							
1	+BT-GB1	Battery with battery management system (BMS on CAN)	Incoe	48 V 50 Ah LFP	48 V DC nominal, 50 Ah, $LiFePO_4$	1	20, 51
2	+EB-QA1	Circuit breaker, 2-pole: main switch (supply disconnecting device)	ABB	S202-C32	2P, 32 A, C characteristic	1	20
3	+EB-FC1	Circuit breaker: feeder, left motor driver	ABB	S201-C13	1P, 13 A, C characteristic	1	20
4	+EB-FC2	Circuit breaker: feeder, right motor driver	ABB	S201-C13	1P, 13 A, C characteristic	1	20
5	+EB-FC3	Circuit breaker: 24 V control bus	ABB	S201-C6	1P, 6 A, C characteristic	1	30
6	+EB-FC4	Circuit breaker: 12 V bus	ABB	S201-C10	1P, 10 A, C characteristic	1	32
7	+EB-TA3	DC/DC converter 48 → 24 V, isolated	Mean Well	SD-150C-24	In 36–72 V DC; out 24 V DC, 6.3 A (150 W)	1	20, 30
8	+EB-TA4	DC/DC converter 48 → 12 V, isolated	Mean Well	SD-100C-12	In 36–72 V DC; out 12 V DC (100 W)	1	20, 32
Drives							
9	+EB-TA1	Motor driver, left: brushless DC, CANopen, safe torque off input	Oriental Motor	BLVD-KRD	24–48 V DC	1	21, 42, 50, 51
10	+EB-TA2	Motor driver, right: brushless DC, CANopen, safe torque off input	Oriental Motor	BLVD-KRD	24–48 V DC	1	21, 42, 50, 51
11	+VH-MA1	Brushless motor with holding brake, left	Oriental Motor	BLMR6400SKM-GFV-B	400 W	1	21
12	+VH-MA2	Brushless motor with holding brake, right	Oriental Motor	BLMR6400SKM-GFV-B	400 W	1	21
13	+VH-MA1	Gearhead, parallel shaft, left	Oriental Motor	GFS6G30FR	Ratio 1:30	1	—
14	+VH-MA2	Gearhead, parallel shaft, right	Oriental Motor	GFS6G30FR	Ratio 1:30	1	—
Safety controller and safety sensors							
15	+EB-KE1	Safety controller, main module (Flexi Soft)	SICK	FX3-CPU000000	24 V DC	1	43
16	+EB-KE1	System plug of -KE1: standard FX3 system plug (holds the configuration)	SICK	FX3-MPL000001	—	1	43
17	+EB-KE2	Safety I/O module (Flexi Soft): 8 inputs, 4 outputs	SICK	FX3-XTIO84002	24 V DC	1	40–44
18	+EB-KE5	Drive monitor module (Flexi Soft): 2 encoder inputs	SICK	FX3-MOC000000	Supplied via the module bus	1	12, 43, 45
19	+VH-BG1	Safety laser scanner, front	SICK	nanoScan3 NANS3-AAAZ30AN1 (1100333)	24 V DC; Ethernet	1	30, 41
20	+VH-BG1	System plug of -BG1, with network connection	SICK	NANSX-AAABAEZZ1	Lead 300 mm with M12 8-pin male (power, safety outputs); lead 250 mm with M12 4-pin D-coded female (Ethernet)	1	41
21	+VH-BG5	Safety encoder, left wheel shaft	SICK	DFS60S-BAOL01024	Sin/cos 1 Vpp, 1 024 periods; 24 V DC from -KE5; 3 m cable	1	12, 45
22	+VH-BG6	Safety encoder, right wheel shaft	SICK	DFS60S-BAOL01024	Sin/cos 1 Vpp, 1 024 periods; 24 V DC from -KE5; 3 m cable	1	12, 45
23	+VH-BG5	M12 plug, 8-pin, on the encoder cable of -BG5	SICK	MOC-Y-CABLE	—	1	45
24	+VH-BG6	M12 plug, 8-pin, on the encoder cable of -BG6	SICK	MOC-Y-CABLE	—	1	45
Operator controls							
25	+FR-SJ1	Emergency stop push-button, red, turn to release, 2 NC	Hanyoung Nux	ARE-3-R-22-R	2 NC	1	40
26	+FR-SJ2	Push-button START, green	Hanyoung Nux	ARF-F-1-10-G	1 NO	1	60
27	+FR-SJ3	Push-button RESET, yellow	Hanyoung Nux	ARF-F-1-20-Y	2 NO	1	40, 60
28	+FR-SJ4	Selector switch MANUAL / AUTO, 2 positions	Hanyoung Nux	ARS-S-2-12	2 NO	1	40, 60

No.	Tag	Component	Manufacturer	Type	Main data	Qty	Diagram pages
29	+PD-SJ5	Push-button MANUAL ARM on the pendant: mushroom, black, latching, turn to release	Hanyoung Nux	HY-PA04	1 NC	1	60
30	+PD-SJ6	Pendant button FORWARD	Hanyoung Nux	HY-PA04	1 NO	1	60
31	+PD-SJ7	Pendant button REVERSE	Hanyoung Nux	HY-PA04	1 NO	1	60
32	+PD-SJ8	Pendant button LEFT	Hanyoung Nux	HY-PA04	1 NO	1	60
33	+PD-SJ9	Pendant button RIGHT	Hanyoung Nux	HY-PA04	1 NO	1	60
34	+FR-PH1	Display (operator pages)	Jawest	JWS080-M-QD	24 V DC	1	31
35	+VH-PJ1	Melody horn, 2 sounds (movement, alarm)	PATLITE	BK-24AE-K	24 V DC	1	30, 61
Vehicle control							
36	+EB-KE3	Industrial PC (vehicle controller), fanless, Intel N97	NEXCOM	NEU-X102-N97	12 V DC; 2 × Ethernet	1	11, 32, 51
37	+EB-KE3	RAM, built into -KE3	V-GEN	DDR4 3200 MHz 8 GB	8 GB	1	—
38	+EB-KE3	Storage, built into -KE3	V-GEN	M.2 SATA 2242 128 GB	128 GB	1	—
39	+EB-KE3	Wi-Fi card, built into -KE3	Intel	AX211NGW	M.2 Wi-Fi / Bluetooth module	1	11
40	+EB-KE4	Digital I/O module, 16 DI / 16 DO, Ethernet (Modbus TCP)	CKDZ-IoT	CK-TP5162	24 V DC	1	30, 60–62
41	+EB-TF1	USB-to-CAN interface, on an IPC USB port	CANable	CANable V2.0S	USB-powered; built-in 120 Ω termination	1	11, 51
Sensors							
42	+VH-BG3	Magnetic line sensor with built-in IMU, CANopen	SICK	MLSE-0200A2NQ0	24 V DC (9–30 V); range 200 mm; lead 0.3 m with M12 4-pin male; no internal terminator	1	31, 51
43	+VH-BG4	UHF RFID reader, Ethernet	Chafon	CF821	12 V DC	1	11, 32
Electrical box							
44	+EB-GQ1	Fan, electrical box	NMB-MAT	3110KL-05W-B60	24 V DC	1	31
45	+EB-GQ2	Fan, electrical box	NMB-MAT	3110KL-05W-B60	24 V DC	1	31
46	+EB-GQ3	Fan, electrical box	NMB-MAT	3110KL-05W-B60	24 V DC	1	31
Mechanical							
47	+VH	Drive wheel	IGP	Custom fabrication	Ø 180 mm; width 50 mm	2	—
48	+VH	Castor	IGP	Swivel castor	Ø 75 mm, swivel	2	—
49	+VH	Towing hook, operated by hand	IGP	Custom fabrication	Coupling height 250 mm; towed mass max. 1 000 kg	1	—

Cables

Cables of circuit diagram IGP-IPR-CD-001.

No.	Tag	Manufacturer	Cable	Cores	From	To	Diagram pages
1	-WD1	First cable	NYAF single cores, black	2 × 4 mm²	-GB1	-QA1	20
2	-WD2	First cable	NYAF single cores, black	2 × 2.5 mm²	-FC1, 1N48	-WD6	21
3	-WD3	First cable	NYAF single cores, black	2 × 2.5 mm²	-FC2, 1N48	-WD7	21
4	-WD4	First cable	NYAF single cores, black	2 × 1.5 mm²	1P48, 1N48	-TA3	20
5	-WD5	First cable	NYAF single cores, black	2 × 1.5 mm²	1P48, 1N48	-TA4	20
6	-WD6	Oriental Motor	LC03D06A (driver power cable)	3 cores	-WD2	-TA1	21
7	-WD7	Oriental Motor	LC03D06A (driver power cable)	3 cores	-WD3	-TA2	21
8	-WD8	Oriental Motor	CCM020B1AAF (motor connection cable)	Motor + encoder	-TA1	-MA1	21
9	-WD9	Oriental Motor	CCM020B1AAF (motor connection cable)	Motor + encoder	-TA2	-MA2	21
10	-WD10	Jawest	Display power cable	2 cores	1P24, 1N24	-PH1	31
11	-WD11	Nexcom	IPC power cable	2 cores	1P12, 1N12	-KE3	32
12	-WD12	Chafon	RFID reader power cable	2 cores	1P12, 1N12	-BG4	32
13	-WG1	SICK	YF2A18-050UA5XLEAX: M12 8-pin female, A-coded → open end, 5 m, PUR	8 × 0.25 mm²	1P24, 1N24; -KE2	-BG1 system plug	30, 41
14	-WG2	SICK	YF2A14-020UB3XLEAX: M12 4-pin female, A-coded → open end, 2 m, PUR	4 cores	1P24, 1N24; CAN bus	-BG3	31, 51
15	-WG3	Belden	9841 CAN cable + GND core	1 × 2 × 0.32 mm² + 1	CAN bus	-TA1	51
16	-WG4	Belden	9841 CAN cable + GND core	1 × 2 × 0.32 mm² + 1	CAN bus	-TA2	51
17	-WG5	Belden	9841 + battery communication cable	1 × 2 × 0.32 mm² + 1	CAN bus	-GB1 BMS	51
18	-WG6	Vention	USB cable	USB	-KE3	-TF1	11, 51
19	-WG7	Vention	Ethernet patch cable	RJ45	-KE3 (enp2s0)	-KE4	11
20	-WG8	Vention	Ethernet patch cable	RJ45	-KE4	-BG4	11
21	-WG9	SICK	YM2D24-020PN1MRJA4: M12 4-pin D-coded male → RJ45, 2 m, Ethernet	4 cores, screened	-KE3 (enp1s0)	-BG1 system plug	11, 41
22	-WG10	Delta	Pendant cable	8 × 0.5 mm²	-KE4; 1P24, 1N24	+PD	60
23	-WG11	SICK	Encoder cable of -BG5, M12 8-pin plug; all 8 wires to -WG13	8 × 0.15 mm², screened	-BG5	-WG13 head B	45
24	-WG12	SICK	Encoder cable of -BG6, M12 8-pin plug; all 8 wires to -WG13	8 × 0.15 mm², screened	-BG6	-WG13 head C	45
25	-WG13	SICK	FX3-MOC Y-cable: Micro D-sub 15-pin → 2 × M12 8-pin female; all encoder wires	Screened	-KE5	-WG11, -WG12	45